Exam Date & Time: 12-Dec-2023 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOV/DEC 2023

Advanced Machine Learning [CSE-5022]

Marks: 50 **Duration: 180 mins.**

PART-A

Answer all the questions.

1) A) A research institute seeks to classify two types of rare birds, Bird A and Bird B, based on certain characteristics. The attributes for classification include Feather Color, Wing Span, Beak Length, and Vocalization. The provided training data includes a mix of these features for both bird types. Develop a decision tree to categorize these birds according to the specified attributes, and subsequently depict this tree visually.

Bird ID	Feather Color	Wing Span	Beak Length	Vocalization	Species (A/B)
1	Black	Wide	Long	Loud	A
2	White	Narrow	Short	Quiet	B
3	Black	Narrow	Long	Loud	A
4	White	Wide	Short	Quiet	B
5	Black	Wide	Short	Loud	A
6	White	Narrow	Long	Quiet	B
7	Black	Narrow	Short	Loud	A
8	White	Wide	Long	Quiet	B
9	Black	Wide	Long	Quiet	A
10	White	Narrow	Short	Loud	B

(5)

B) Apply the Naïve Bayes classification method to a dataset containing information about email messages. The dataset includes various attributes of each email. Your task is to classify a new email based on the following characteristics: Length - Short, Keyword - "Discount", and Sender - Known. Use the Naïve Bayes classifier to determine the likelihood of this email being spam.

(5)

Email ID	Length	Keyword	Sender	Spam (Yes/No)
1	Long	"Discount"	Unknown	Yes
2	Short	"Meeting"	Known	No
3	Long	"Free"	Unknown	Yes
4	Short	"Discount"	Known	No
5	Long	"Meeting"	Unknown	No
6	Short	"Free"	Known	Yes
7	Long	"Discount"	Known	No
8	Short	"Meeting"	Unknown	No
9	Long	"Free"	Known	Yes
10	Short	"Discount"	Unknown	Yes

2) A) Examine the principles of ensemble learning and the composition of an ensemble model. Also, identify the technical difficulties associated with constructing a model based on a single estimator. (3)

B) Outline the process of feature selection and detail the various methodologies employed in choosing features for data modeling. (3)

C) Determine the Manhattan distance and Cosine Similarity for the provided test dataset X(1.4,1.6).

	A_1	A_2
1	1.5	1.7
2	2	1.9

(4)

3) A) Subdivide the following customers into two groups using the K-Means clustering algorithm.

sr_no	age	amount
C1	20	500
C2	40	1000
C3	30	800
C4	18	300
C5	28	1200
C6	35	1400
C7	45	1800

(5)

B) Demonstrate backpropagation and its foundational role in neural network training. Additionally, detail the mathematical principles underlying the backpropagation Algorithm. (5)

4) Illustrate the concept of a genetic algorithm and describe its operational mechanism. (5)

A)

B) Determine the Perceptron training rule for AND gate. W1=1.4, W2=0.8 Threshold =0.75 and Learning rate n=1. (if output is greater than threshold value considers it 1 else 0) (5)

5) Apply the KSOM to cluster a set of given vectors. The vectors to be clustered are [1100], [0011], and [1111]. You are required to form 2 clusters. Assume the initial learning rate is 0.6 and the initial weights are w1= [0.2,0.8,0.4,0.6] and w2= [0.7,0.3,0.5,0.1]. (5)

A)

B) Differentiate between backward chaining and forward chaining with example (5)

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